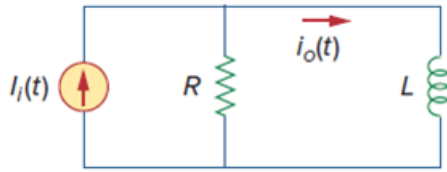


## Problem

Find the transfer function  $I_o/I_i$  of the  $RL$  circuit in Fig. 14.68. Express it using  $\omega_0 = R/L$ .

**Figure 14.68:**



## Step-by-step solution

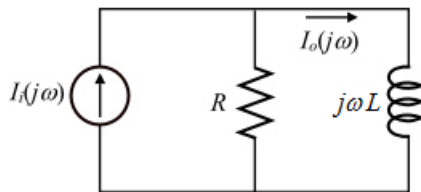
## Step 1 of 3

Refer to circuit diagram in Figure 14.68 in the textbook.

In frequency domain inductor is represented as,

$$X_L = j\omega L$$

Draw the circuit diagram in frequency domain.



**Figure 1**

## Step 2 of 3

Determine the transfer function using current division principle.

$$I_o(j\omega) = \left( \frac{R}{R + j\omega L} \right) I_i(j\omega)$$

$$\frac{I_o(j\omega)}{I_i(j\omega)} = \frac{R}{R + j\omega L}$$

Therefore, the transfer function,  $I_o / I_i$  of the  $RL$  circuit is  $\boxed{\frac{R}{R + j\omega L}}$ .

## Step 3 of 3

Express the obtained transfer function using  $\omega_0$ .

The angular frequency,  $\omega_0$  is,

$$\omega_0 = \frac{R}{L}$$

Re write the transfer function  $I_o / I_i$  as follows:

$$\begin{aligned} \frac{I_o(j\omega)}{I_i(j\omega)} &= \frac{R}{R + j\omega L} \\ &= \frac{1}{1 + j \frac{\omega L}{R}} \\ &= \frac{1}{1 + j \frac{\omega}{\omega_0}} \quad \left( \text{Here } \omega_0 = \frac{R}{L} \right) \end{aligned}$$

Therefore, the transfer function,  $I_o / I_i$  in terms of  $\omega_0$  is

$$\boxed{\frac{1}{1 + j \frac{\omega}{\omega_0}}}.$$